Ex. No.:2	AI-Powered Blog Content Generator	URK23CS1167
13.08.2026		

Aim:

To build an AI-powered Blog Content Generator using Ollama that generates blogs, titles, summaries, SEO keywords, and meta descriptions based on user prompts.

Problem Statement:

Design and develop a modular AI-powered Blog Content Generator using Ollama. The application should generate high-quality blogs from user-provided topics, writing styles, and word counts. It should also generate blog titles, summaries, SEO keywords, and meta descriptions while intelligently routing user requests using an LLM-based intent detector.

Software Requirements:

Software	Version
Python	3.13.18
Ollama	0.6.2
VS Code	1.133.0
Pandas	3.0.5
Sentence Transformers	5.7.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Input Parameters:

Parameters	Description
Blog Topic	Subject of the blog. Example: Electric Vehicles in India
Writing Style	Professional, Informative, Conversational, Storytelling, Technical or Listicle
Tone	Friendly, Formal, Persuasive, Inspirational, Neutral or Humorous
Word count	Target length of the blog (150 – 1500 words). Example: 400
Target Audience	Who the blog is written for. Example: Engineering students
Additional Instructions	Optional extra guidance. Example: Include benefits, challenges and future

System Architecture:

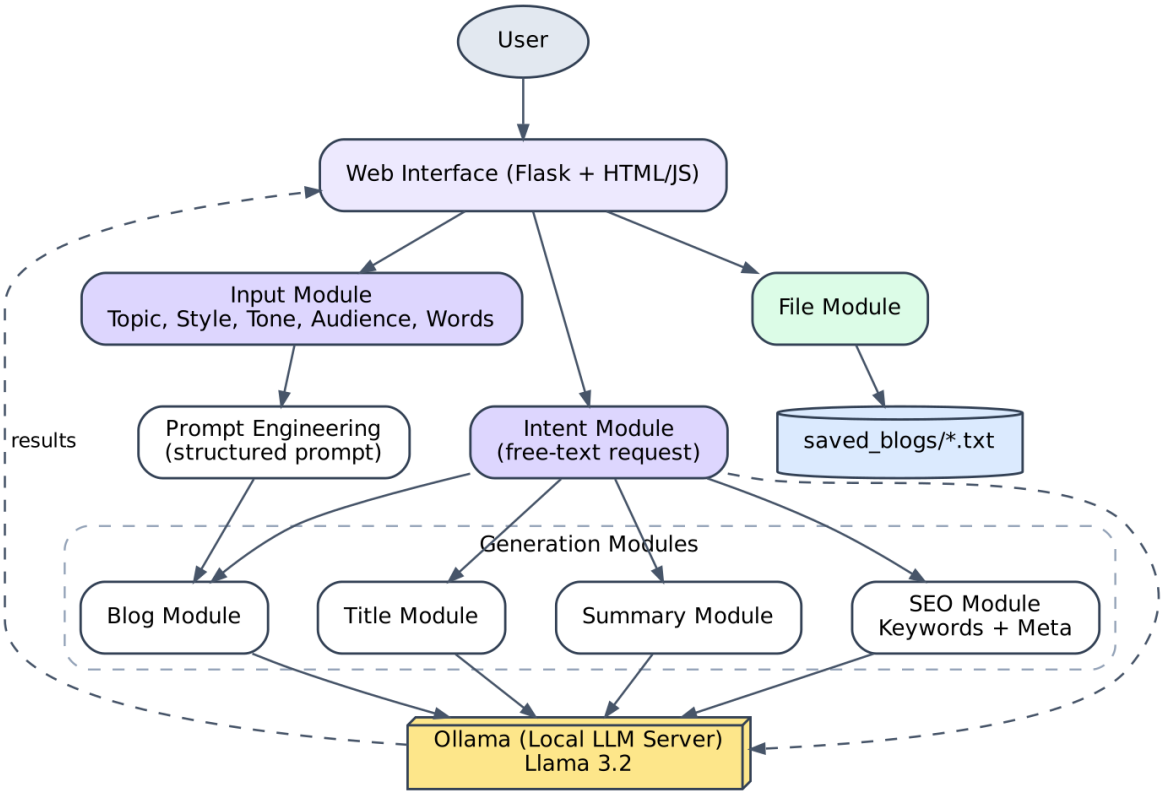


Fig. 2.1 System architecture

Module Design:

Module	Responsibility
Input Module	Collects and validates topic, writing style, tone, audience, word count and additional instructions from the web form.
Blog Module	Builds a structured prompt (prompt engineering) and generates the blog with title, Introduction, main sections and Conclusion using Llama 3.2.
Title Module	Generates five catchy, SEO-friendly blog titles under 70 characters.
Summary Module	Generates a concise summary of about 60 words for the generated blog.
SEO Module	Extracts 8 SEO keywords (short and long-tail) and writes a meta description within 160 characters with a search-result preview.
Intent Module	Understands a free-text request (e.g. 'suggest catchy headings') and routes it to the Blog, Title, Summary, SEO, Meta or Save action.
File Module	Saves the blog with its titles, summary, keywords and meta description to a .txt file in the saved_blogs folder.

Implementation:

• GitHub Link - https://github.com/MONICK10/GPT4-Exp2-Blog-Content-Generator

• LinkedIn Post Link - <add LinkedIn post link in config.json>
• Run: `python app.py` → `http://localhost:5002`

Output Screenshots:

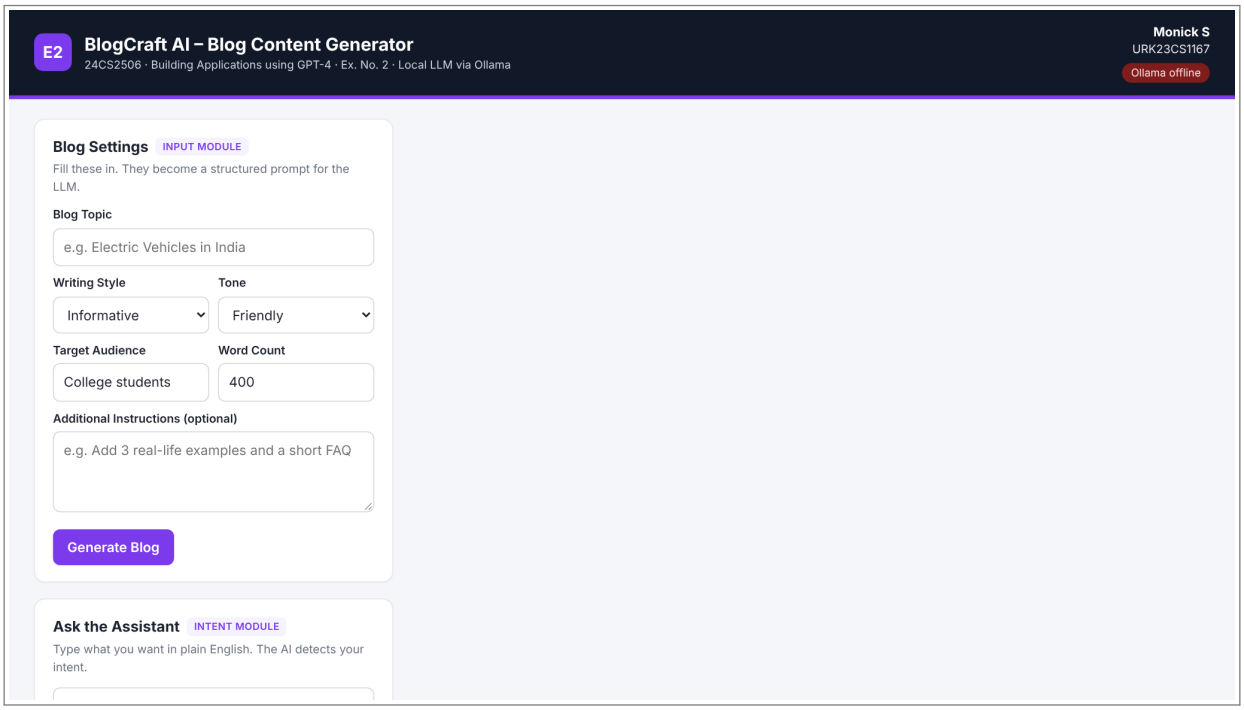


Fig. 2.2 Home Screen

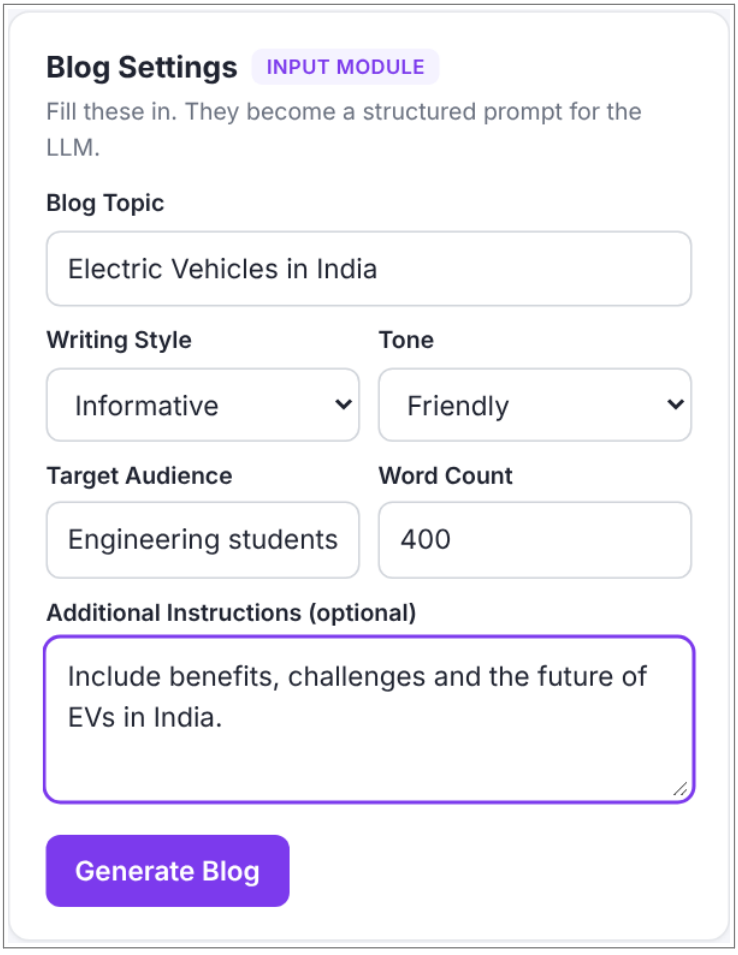


Fig. 2.3 User Input

[Draft] The AI output screenshots are added automatically when you run python tools/capture.py 2 on your laptop with Ollama.

Result:

Thus, an AI-powered Blog Content Generator was successfully developed using Ollama. The application generates blogs based on user prompts, creates titles, summaries, SEO keywords, and meta descriptions, saves the generated content to a text file, and demonstrates the use of local Large Language Models (LLMs) for content generation without requiring cloud-based APIs.